

Adult Table II: IID — Same-checkpoint round70 correction candidate

Historical label	Metric	Benign	F-Flip	FedSA	S-DFA	Sp-DFA
FedAvg	ACC (%)	83.14 ¹	83.15 ± 0.66	76.45 ± 1.68	75.89 ¹	79.55 ¹
	AEOD	0.0740 ¹	0.0443 ± 0.0442	0.2578 ± 0.0749	0.0039 ¹	0.0926 ¹
	ASPD	0.1148 ¹	0.1369 ± 0.0197	0.0466 ± 0.0337	0.0046 ¹	0.0802 ¹
FairFed	ACC (%)	83.21 ¹	83.06 ± 0.63	73.85 ± 4.07	75.91 ¹	79.15 ¹
	AEOD	0.0723 ¹	0.0448 ± 0.0459	0.3105 ± 0.0887	0.0027 ¹	0.0719 ¹
	ASPD	0.1146 ¹	0.1337 ± 0.0187	0.0865 ± 0.0675	0.0046 ¹	0.0671 ¹
LoGoFair	ACC (%)	82.12 ¹	81.71 ± 0.87	75.52 ± 0.05	75.53 ¹	78.31 ¹
	AEOD	0.0135 ¹	0.0342 ± 0.0325	0.0069 ± 0.0079	0.0227 ¹	0.0264 ¹
	ASPD	0.0749 ¹	0.0669 ± 0.0131	0.0011 ± 0.0008	0.0024 ¹	0.0215 ¹
FedWA [†]	ACC (%)	83.11 ¹	82.77 ¹	83.46 ± 0.60	82.84 ¹	83.00 ¹
	AEOD	0.0920 ¹	0.0470 ¹	0.0727 ± 0.0485	0.0763 ¹	0.0579 ¹
	ASPD	0.1064 ¹	0.1345 ¹	0.1170 ± 0.0237	0.1127 ¹	0.1245 ¹
LASA	ACC (%)	82.85 ¹	83.13 ± 0.55	82.66 ± 0.89	82.93 ¹	82.85 ¹
	AEOD	0.0860 ¹	0.0493 ± 0.0481	0.0624 ± 0.0407	0.0676 ¹	0.0653 ¹
	ASPD	0.1090 ¹	0.1301 ± 0.0137	0.0853 ± 0.0212	0.1172 ¹	0.1181 ¹
FedAA	ACC (%)	83.58 ¹	83.44 ± 0.35	83.26 ± 0.52	83.31 ¹	83.37 ¹
	AEOD	0.0947 ¹	0.0627 ± 0.0662	0.0548 ± 0.0519	0.0359 ¹	0.0688 ¹
	ASPD	0.1011 ¹	0.1166 ± 0.0228	0.1092 ± 0.0181	0.1291 ¹	0.1128 ¹
Median	ACC (%)	83.41 ¹	82.98 ± 0.66	81.75 ± 1.25	83.77 ¹	83.73 ¹
	AEOD	0.0461 ¹	0.0595 ± 0.0367	0.0427 ± 0.0297	0.0786 ¹	0.0580 ¹
	ASPD	0.1192 ¹	0.1275 ± 0.0270	0.0793 ± 0.0315	0.1063 ¹	0.1240 ¹
FLTrust	ACC (%)	82.12 ¹	83.33 ± 0.53	82.51 ± 2.58	82.99 ¹	83.08 ¹
	AEOD	0.2334 ¹	0.0785 ± 0.0840	0.0515 ± 0.0375	0.0248 ¹	0.1533 ¹
	ASPD	0.0207 ¹	0.1525 ± 0.0398	0.1166 ± 0.0294	0.1231 ¹	0.0675 ¹
Huber-BRFL	ACC (%)	83.09 ¹	83.10 ± 0.66	81.48 ± 0.65	83.54 ¹	83.19 ¹
	AEOD	0.0670 ¹	0.0429 ± 0.0476	0.0897 ± 0.0436	0.0601 ¹	0.0403 ¹
	ASPD	0.1159 ¹	0.1374 ± 0.0171	0.0608 ± 0.0195	0.1098 ¹	0.1335 ¹
FLGMM	ACC (%)	82.30 ¹	82.89 ± 0.65	82.30 ± 1.00	82.58 ¹	81.67 ¹
	AEOD	0.0671 ¹	0.0469 ± 0.0296	0.0486 ± 0.0442	0.0769 ¹	0.1581 ¹
	ASPD	0.1242 ¹	0.1205 ± 0.0192	0.1083 ± 0.0219	0.1163 ¹	0.0492 ¹
SmartFL	ACC (%)	83.17 ¹	83.22 ± 0.58	81.59 ± 0.91	80.61 ¹	82.89 ¹
	AEOD	0.0632 ¹	0.0541 ± 0.0454	0.1267 ± 0.0616	0.0188 ¹	0.0103 ¹
	ASPD	0.1193 ¹	0.1299 ± 0.0191	0.0519 ± 0.0287	0.0607 ¹	0.1606 ¹
Fed-NGA	ACC (%)	82.42 ¹	82.03 ± 2.01	79.09 ± 4.76	82.02 ¹	82.75 ¹
	AEOD	0.1208 ¹	0.0723 ± 0.0629	0.1079 ± 0.0592	0.1502 ¹	0.0927 ¹
	ASPD	0.0664 ¹	0.1370 ± 0.0701	0.0441 ± 0.0300	0.0516 ¹	0.0703 ¹
FedDNA	ACC (%)	82.99 ¹	83.16 ± 0.63	82.94 ± 0.62	83.09 ¹	82.77 ¹
	AEOD	0.0490 ¹	0.0584 ± 0.0501	0.0628 ± 0.0354	0.0612 ¹	0.0594 ¹
	ASPD	0.1239 ¹	0.1321 ± 0.0254	0.1073 ± 0.0220	0.1186 ¹	0.1194 ¹
FairGuard	ACC (%)	83.06 ¹	82.93 ± 0.70	53.94 ± 26.05	75.43 ¹	78.65 ¹
	AEOD	0.0525 ¹	0.0355 ± 0.0154	0.0344 ± 0.0999	0.0000 ¹	0.0587 ¹
	ASPD	0.1207 ¹	0.1182 ± 0.0207	0.0357 ± 0.0794	0.0000 ¹	0.0549 ¹
FairGuard → FLTrust	ACC (%)	82.89 ¹	81.49 ± 5.07	73.00 ± 18.07	82.36 ¹	81.31 ¹
	AEOD	0.0703 ¹	0.0319 ± 0.0138	0.0762 ± 0.0662	0.0145 ¹	0.0520 ¹
	ASPD	0.0913 ¹	0.1385 ± 0.0377	0.0969 ± 0.0603	0.1455 ¹	0.1955 ¹
Cosine + fairness [‡]	ACC (%)	83.11 ¹	78.49 ± 9.61	82.56 ± 0.76	82.08 ¹	82.22 ¹
	AEOD	0.0239 ¹	0.0436 ± 0.0320	0.0342 ± 0.0254	0.0095 ¹	0.0278 ¹
	ASPD	0.0931 ¹	0.1313 ± 0.0442	0.1023 ± 0.0253	0.0699 ¹	0.0658 ¹

Superscript 1: single seed, SD unavailable. Other cells: mean ± sample SD (ddof=1), n=10 except FedWA/FedSA n=3. No accuracy-based suppression of fairness values. AEOD denotes the implemented absolute TPR gap.

[†] FedWA is a historical display label whose numeric source is AdaAggRL. [‡] The Cosine row combines GuardFed-AD2 in Benign/S-DFA/Sp-DFA with GuardFed in F-Flip/FedSA. These are source identifications, not certifications of the named baseline algorithms.

Every metric uses the same final-round record within a seed. Mixed historical training recipes and sample sizes remain disclosed; n=1 has no repeated-seed uncertainty. This candidate removes checkpoint-selection mixing but does not resolve baseline implementation fidelity or create missing seeds.

Adult Table II: non-IID — Same-checkpoint round70 correction candidate

Historical label	Metric	Benign	F-Flip	FedSA	S-DFA	Sp-DFA
FedAvg	ACC (%)	83.83 ¹	83.35 ± 0.34	71.45 ± 11.25	76.52 ¹	80.07 ¹
	AEOD	0.0252 ¹	0.0467 ± 0.0380	0.2664 ± 0.1953	0.0067 ¹	0.0787 ¹
	ASPD	0.1231 ¹	0.1548 ± 0.0202	0.1195 ± 0.1295	0.0119 ¹	0.0730 ¹
FairFed	ACC (%)	83.76 ¹	83.22 ± 0.42	72.33 ± 10.40	75.98 ¹	79.19 ¹
	AEOD	0.0243 ¹	0.0483 ± 0.0500	0.3027 ± 0.1296	0.0020 ¹	0.0895 ¹
	ASPD	0.1290 ¹	0.1469 ± 0.0104	0.1076 ± 0.1264	0.0051 ¹	0.0618 ¹
LoGoFair	ACC (%)	82.35 ¹	81.73 ± 0.97	77.07 ± 1.76	77.79 ¹	79.62 ¹
	AEOD	0.0443 ¹	0.0366 ± 0.0305	0.0124 ± 0.0142	0.0052 ¹	0.0347 ¹
	ASPD	0.0704 ¹	0.0670 ± 0.0129	0.0158 ± 0.0177	0.0241 ¹	0.0363 ¹
FedWA [†]	ACC (%)	83.64 ¹	83.68 ¹	83.39 ± 0.14	83.49 ¹	83.75 ¹
	AEOD	0.0146 ¹	0.0062 ¹	0.0613 ± 0.0355	0.0180 ¹	0.0092 ¹
	ASPD	0.1339 ¹	0.1401 ¹	0.1087 ± 0.0087	0.1282 ¹	0.1393 ¹
LASA	ACC (%)	83.68 ¹	83.25 ± 0.34	82.50 ± 0.87	83.60 ¹	83.94 ¹
	AEOD	0.0417 ¹	0.0502 ± 0.0596	0.0713 ± 0.0396	0.0241 ¹	0.0091 ¹
	ASPD	0.1264 ¹	0.1422 ± 0.0138	0.0876 ± 0.0217	0.1232 ¹	0.1382 ¹
FedAA	ACC (%)	83.33 ¹	83.02 ± 0.55	83.07 ± 0.79	83.98 ¹	83.75 ¹
	AEOD	0.0183 ¹	0.0286 ± 0.0235	0.0376 ± 0.0319	0.0020 ¹	0.0547 ¹
	ASPD	0.1136 ¹	0.1129 ± 0.0192	0.1118 ± 0.0170	0.1200 ¹	0.1222 ¹
Median	ACC (%)	82.82 ¹	83.08 ± 0.60	82.24 ± 0.77	82.63 ¹	83.27 ¹
	AEOD	0.0701 ¹	0.0452 ± 0.0407	0.0402 ± 0.0302	0.0181 ¹	0.0451 ¹
	ASPD	0.0796 ¹	0.1338 ± 0.0210	0.0904 ± 0.0164	0.0815 ¹	0.0972 ¹
FLTrust	ACC (%)	83.52 ¹	82.73 ± 0.98	82.80 ± 1.77	83.58 ¹	83.58 ¹
	AEOD	0.0151 ¹	0.0908 ± 0.0697	0.0572 ± 0.0468	0.0003 ¹	0.0126 ¹
	ASPD	0.1033 ¹	0.1722 ± 0.0501	0.1167 ± 0.0338	0.1228 ¹	0.1486 ¹
Huber-BRFL	ACC (%)	83.68 ¹	83.19 ± 0.41	81.13 ± 0.74	83.98 ¹	83.98 ¹
	AEOD	0.0175 ¹	0.0524 ± 0.0483	0.0975 ± 0.0354	0.0321 ¹	0.0116 ¹
	ASPD	0.1295 ¹	0.1471 ± 0.0114	0.0542 ± 0.0176	0.1253 ¹	0.1420 ¹
FLGMM	ACC (%)	81.45 ¹	81.74 ± 0.70	81.61 ± 0.94	81.65 ¹	81.38 ¹
	AEOD	0.0994 ¹	0.0539 ± 0.0246	0.0531 ± 0.0267	0.0752 ¹	0.0801 ¹
	ASPD	0.0614 ¹	0.0923 ± 0.0298	0.0936 ± 0.0344	0.0660 ¹	0.0514 ¹
SmartFL	ACC (%)	83.68 ¹	83.06 ± 0.40	81.62 ± 0.89	81.44 ¹	83.72 ¹
	AEOD	0.0199 ¹	0.0603 ± 0.0461	0.1148 ± 0.0540	0.0418 ¹	0.0100 ¹
	ASPD	0.1277 ¹	0.1512 ± 0.0163	0.0579 ± 0.0295	0.0817 ¹	0.1466 ¹
Fed-NGA	ACC (%)	83.09 ¹	82.65 ± 0.47	80.38 ± 3.39	81.25 ¹	81.85 ¹
	AEOD	0.0525 ¹	0.0596 ± 0.0451	0.1009 ± 0.0447	0.0596 ¹	0.0911 ¹
	ASPD	0.0981 ¹	0.1237 ± 0.0415	0.0648 ± 0.0403	0.0666 ¹	0.0666 ¹
FedDNA	ACC (%)	83.45 ¹	82.97 ± 0.59	83.06 ± 0.74	83.66 ¹	83.66 ¹
	AEOD	0.0127 ¹	0.0626 ± 0.0763	0.0479 ± 0.0479	0.0349 ¹	0.0025 ¹
	ASPD	0.1271 ¹	0.1442 ± 0.0225	0.1174 ± 0.0234	0.1259 ¹	0.1401 ¹
FairGuard	ACC (%)	83.86 ¹	83.10 ± 0.60	61.03 ± 25.49	75.43 ¹	82.65 ¹
	AEOD	0.0047 ¹	0.0311 ± 0.0245	0.0302 ± 0.0775	0.0000 ¹	0.0014 ¹
	ASPD	0.1310 ¹	0.1186 ± 0.0204	0.0342 ± 0.0514	0.0000 ¹	0.0839 ¹
FairGuard → FLTrust	ACC (%)	83.88 ¹	81.39 ± 3.65	80.19 ± 7.67	83.75 ¹	83.87 ¹
	AEOD	0.0036 ¹	0.0497 ± 0.0505	0.0403 ± 0.0329	0.0308 ¹	0.0074 ¹
	ASPD	0.1154 ¹	0.1493 ± 0.0479	0.0942 ± 0.0345	0.1565 ¹	0.1231 ¹
Cosine + fairness [‡]	ACC (%)	81.84 ¹	80.50 ± 5.79	81.92 ± 2.10	82.11 ¹	81.99 ¹
	AEOD	0.0268 ¹	0.0430 ± 0.0237	0.0577 ± 0.0272	0.0418 ¹	0.0307 ¹
	ASPD	0.0638 ¹	0.1060 ± 0.0444	0.1125 ± 0.0362	0.0664 ¹	0.0666 ¹

Superscript 1: single seed, SD unavailable. Other cells: mean ± sample SD (ddof=1), n=10 except FedWA/FedSA n=3. No accuracy-based suppression of fairness values. AEOD denotes the implemented absolute TPR gap.

[†] FedWA is a historical display label whose numeric source is AdaAggRL. [‡] The Cosine row combines GuardFed-AD2 in Benign/S-DFA/Sp-DFA with GuardFed in F-Flip/FedSA. These are source identifications, not certifications of the named baseline algorithms.

Every metric uses the same final-round record within a seed. Mixed historical training recipes and sample sizes remain disclosed; n=1 has no repeated-seed uncertainty. This candidate removes checkpoint-selection mixing but does not resolve baseline implementation fidelity or create missing seeds.

Adult Table II: IID — Historical selections reconstructed

Historical label	Metric	Benign	F-Flip	FedSA	S-DFA	Sp-DFA
FedAvg	ACC (%)	83.18 ¹	83.00 ± 0.66	76.07 ± 1.79	75.89 ¹	79.59 ¹
	AEOD	0.0615 ¹	0.0559 ± 0.0420	0.2667 ± 0.0806	0.0039 ¹	0.0917 ¹
	ASPD	0.1098 ¹	0.1367 ± 0.0192	0.0498 ± 0.0402	0.0044 ¹	0.0798 ¹
FairFed	ACC (%)	83.26 ¹	82.95 ± 0.72	73.55 ± 3.97	75.91 ¹	79.19 ¹
	AEOD	0.0570 ¹	0.0343 ± 0.0404	0.3225 ± 0.0881	0.0021 ¹	0.0714 ¹
	ASPD	0.1137 ¹	0.1296 ± 0.0202	0.0948 ± 0.0650	0.0044 ¹	0.0671 ¹
LoGoFair	ACC (%)	82.12 ¹	81.59 ± 0.95	75.53 ± 0.05	75.53 ¹	78.33 ¹
	AEOD	0.0107 ¹	0.0457 ± 0.0274	0.0075 ± 0.0082	0.0227 ¹	0.0248 ¹
	ASPD	0.0711 ¹	0.0656 ± 0.0167	0.0011 ± 0.0008	0.0024 ¹	0.0215 ¹
FedWA [†]	ACC (%)	83.11 ¹	82.77 ¹	83.43 ± 0.69	82.84 ¹	83.03 ¹
	AEOD	0.0698 ¹	0.0195 ¹	0.0635 ± 0.0461	0.0613 ¹	0.0379 ¹
	ASPD	0.1043 ¹	0.1314 ¹	0.1174 ± 0.0191	0.1127 ¹	0.1241 ¹
LASA	ACC (%)	82.93 ¹	83.06 ± 0.68	82.69 ± 0.88	82.93 ¹	82.93 ¹
	AEOD	0.0695 ¹	0.0638 ± 0.0500	0.0617 ± 0.0412	0.0571 ¹	0.0416 ¹
	ASPD	0.1089 ¹	0.1360 ± 0.0204	0.0867 ± 0.0212	0.1158 ¹	0.1181 ¹
FedAA	ACC (%)	83.59 ¹	83.38 ± 0.51	83.35 ± 0.45	83.49 ¹	83.44 ¹
	AEOD	0.0660 ¹	0.0886 ± 0.0755	0.0499 ± 0.0580	0.0359 ¹	0.0591 ¹
	ASPD	0.0878 ¹	0.1245 ± 0.0302	0.1152 ± 0.0205	0.1021 ¹	0.0948 ¹
Median	ACC (%)	83.41 ¹	82.97 ± 0.69	81.85 ± 1.24	83.80 ¹	83.73 ¹
	AEOD	0.0287 ¹	0.0660 ± 0.0362	0.0430 ± 0.0335	0.0498 ¹	0.0396 ¹
	ASPD	0.1140 ¹	0.1263 ± 0.0282	0.0777 ± 0.0303	0.1063 ¹	0.1210 ¹
FLTrust	ACC (%)	83.13 ¹	82.44 ± 0.82	83.13 ± 1.25	83.48 ¹	83.22 ¹
	AEOD	0.0820 ¹	0.1365 ± 0.0802	0.0436 ± 0.0280	0.0013 ¹	0.1207 ¹
	ASPD	0.0207 ¹	0.2157 ± 0.0403	0.1126 ± 0.0339	0.0774 ¹	0.0598 ¹
Huber-BRFL	ACC (%)	83.17 ¹	82.95 ± 0.67	81.55 ± 0.66	83.68 ¹	83.19 ¹
	AEOD	0.0627 ¹	0.0543 ± 0.0441	0.0886 ± 0.0440	0.0290 ¹	0.0331 ¹
	ASPD	0.1101 ¹	0.1362 ± 0.0184	0.0569 ± 0.0165	0.1098 ¹	0.1313 ¹
FLGMM	ACC (%)	83.19 ¹	82.40 ± 0.64	82.54 ± 0.94	82.69 ¹	81.82 ¹
	AEOD	0.0561 ¹	0.0627 ± 0.0310	0.0419 ± 0.0397	0.0607 ¹	0.1117 ¹
	ASPD	0.1022 ¹	0.1288 ± 0.0307	0.1079 ± 0.0226	0.0892 ¹	0.0492 ¹
SmartFL	ACC (%)	83.23 ¹	82.94 ± 0.53	82.04 ± 0.72	80.61 ¹	83.02 ¹
	AEOD	0.0556 ¹	0.0685 ± 0.0432	0.0640 ± 0.0432	0.0041 ¹	0.0081 ¹
	ASPD	0.1144 ¹	0.1462 ± 0.0350	0.0790 ± 0.0198	0.0564 ¹	0.1040 ¹
Fed-NGA	ACC (%)	82.60 ¹	81.81 ± 2.14	81.07 ± 1.88	83.25 ¹	82.75 ¹
	AEOD	0.1042 ¹	0.0915 ± 0.0552	0.0858 ± 0.0441	0.0764 ¹	0.0356 ¹
	ASPD	0.0531 ¹	0.1452 ± 0.0728	0.0650 ± 0.0568	0.0098 ¹	0.0664 ¹
FedDNA	ACC (%)	82.99 ¹	82.89 ± 0.69	83.03 ± 0.65	83.15 ¹	83.08 ¹
	AEOD	0.0320 ¹	0.0721 ± 0.0572	0.0617 ± 0.0378	0.0565 ¹	0.0441 ¹
	ASPD	0.1180 ¹	0.1482 ± 0.0266	0.1101 ± 0.0220	0.1154 ¹	0.1155 ¹
FairGuard	ACC (%)	83.30 ¹	82.85 ± 0.84	54.13 ± 26.10	75.43 ¹	78.70 ¹
	AEOD	0.0232 ¹	0.0219 ± 0.0160	0.0362 ± 0.1072	0.0000 ¹	0.0498 ¹
	ASPD	0.1154 ¹	0.1125 ± 0.0251	0.0347 ± 0.0774	0.0000 ¹	0.0469 ¹
FairGuard → FLTrust	ACC (%)	82.99 ¹	82.37 ± 1.72	75.85 ± 17.79	84.14 ¹	82.86 ¹
	AEOD	0.0011 ¹	0.0115 ± 0.0069	0.0241 ± 0.0284	0.0035 ¹	0.0060 ¹
	ASPD	0.0584 ¹	0.0959 ± 0.0353	0.0790 ± 0.0594	0.1084 ¹	0.0788 ¹
Cosine + fairness [‡]	ACC (%)	83.11 ¹	81.90 ± 1.74	83.10 ± 0.55	82.21 ¹	82.22 ¹
	AEOD	0.0003 ¹	0.0264 ± 0.0221	0.0475 ± 0.0442	0.0038 ¹	0.0094 ¹
	ASPD	0.0639 ¹	0.0850 ± 0.0317	0.1190 ± 0.0160	0.0629 ¹	0.0592 ¹

Superscript 1: single seed, SD unavailable. Other cells: mean ± sample SD (ddof=1), n=10 except FedWA/FedSA n=3. No accuracy-based suppression of fairness values. AEOD denotes the implemented absolute TPR gap.

[†] FedWA is a historical display label whose numeric source is AdaAggRL. [‡] The Cosine row combines GuardFed-AD2 in Benign/S-DFA/Sp-DFA with GuardFed in F-Flip/FedSA. These are source identifications, not certifications of the named baseline algorithms.

Historical n=1 cells independently maximize ACC and minimize AEOD/ASPD over rounds61–70; 94/98 conditions have no common selected round. F-Flip/FedSA n=10 use opposite selection directions for different method groups. FedWA/FedSA n=3 jointly select ACC-.5(AEOD+ASPD). These test-selected results do not provide an unbiased matched-method ranking.

FairGuard IID FedSA ACC is restored as54.13% from the source, versus59.13% in the submitted PDF. Original submission retained separately.

Adult Table II: non-IID — Historical selections reconstructed

Historical label	Metric	Benign	F-Flip	FedSA	S-DFA	Sp-DFA
FedAvg	ACC (%)	83.83 ¹	83.12 ± 0.40	70.17 ± 11.53	76.52 ¹	80.10 ¹
	AEOD	0.0206 ¹	0.0715 ± 0.0494	0.2768 ± 0.1961	0.0048 ¹	0.0763 ¹
	ASPD	0.1223 ¹	0.1547 ± 0.0201	0.1297 ± 0.1340	0.0113 ¹	0.0717 ¹
FairFed	ACC (%)	83.80 ¹	83.10 ± 0.57	71.90 ± 10.43	75.99 ¹	79.19 ¹
	AEOD	0.0119 ¹	0.0466 ± 0.0506	0.3087 ± 0.1314	0.0014 ¹	0.0880 ¹
	ASPD	0.1277 ¹	0.1417 ± 0.0129	0.1106 ± 0.1300	0.0049 ¹	0.0611 ¹
LoGoFair	ACC (%)	82.37 ¹	81.75 ± 0.84	77.38 ± 2.09	77.79 ¹	79.65 ¹
	AEOD	0.0172 ¹	0.0706 ± 0.0339	0.0223 ± 0.0387	0.0045 ¹	0.0347 ¹
	ASPD	0.0695 ¹	0.0618 ± 0.0184	0.0185 ± 0.0207	0.0237 ¹	0.0268 ¹
FedWA [†]	ACC (%)	83.66 ¹	83.68 ¹	83.39 ± 0.27	83.54 ¹	83.75 ¹
	AEOD	0.0047 ¹	0.0062 ¹	0.0368 ± 0.0197	0.0097 ¹	0.0092 ¹
	ASPD	0.1314 ¹	0.1400 ¹	0.1087 ± 0.0093	0.1227 ¹	0.1393 ¹
LASA	ACC (%)	83.77 ¹	83.09 ± 0.36	82.63 ± 0.84	83.65 ¹	83.94 ¹
	AEOD	0.0186 ¹	0.0627 ± 0.0617	0.0620 ± 0.0381	0.0228 ¹	0.0024 ¹
	ASPD	0.1237 ¹	0.1427 ± 0.0151	0.0899 ± 0.0207	0.1156 ¹	0.1259 ¹
FedAA	ACC (%)	83.33 ¹	82.88 ± 0.64	83.18 ± 0.79	84.00 ¹	83.93 ¹
	AEOD	0.0005 ¹	0.0612 ± 0.0381	0.0382 ± 0.0341	0.0020 ¹	0.0032 ¹
	ASPD	0.1052 ¹	0.1135 ± 0.0236	0.1161 ± 0.0157	0.1200 ¹	0.1161 ¹
Median	ACC (%)	82.82 ¹	82.95 ± 0.65	82.24 ± 0.77	82.67 ¹	83.27 ¹
	AEOD	0.0701 ¹	0.0561 ± 0.0437	0.0402 ± 0.0302	0.0001 ¹	0.0233 ¹
	ASPD	0.0560 ¹	0.1326 ± 0.0207	0.0904 ± 0.0164	0.0725 ¹	0.0836 ¹
FLTrust	ACC (%)	83.82 ¹	81.36 ± 1.51	83.11 ± 1.42	83.76 ¹	84.01 ¹
	AEOD	0.0038 ¹	0.1463 ± 0.0731	0.0617 ± 0.0718	0.0003 ¹	0.0031 ¹
	ASPD	0.0801 ¹	0.2402 ± 0.0546	0.1224 ± 0.0452	0.0796 ¹	0.1380 ¹
Huber-BRFL	ACC (%)	83.70 ¹	83.05 ± 0.51	81.19 ± 0.77	83.98 ¹	83.98 ¹
	AEOD	0.0139 ¹	0.0619 ± 0.0513	0.0886 ± 0.0397	0.0271 ¹	0.0098 ¹
	ASPD	0.1278 ¹	0.1458 ± 0.0113	0.0567 ± 0.0182	0.1181 ¹	0.1370 ¹
FLGMM	ACC (%)	81.45 ¹	81.62 ± 0.92	81.83 ± 0.94	81.78 ¹	81.75 ¹
	AEOD	0.0786 ¹	0.0674 ± 0.0242	0.0519 ± 0.0347	0.0291 ¹	0.0510 ¹
	ASPD	0.0473 ¹	0.1124 ± 0.0275	0.0857 ± 0.0273	0.0485 ¹	0.0469 ¹
SmartFL	ACC (%)	83.74 ¹	82.97 ± 0.61	82.23 ± 0.61	81.44 ¹	83.79 ¹
	AEOD	0.0162 ¹	0.0682 ± 0.0520	0.0621 ± 0.0465	0.0418 ¹	0.0033 ¹
	ASPD	0.1195 ¹	0.1545 ± 0.0162	0.0842 ± 0.0161	0.0713 ¹	0.1326 ¹
Fed-NGA	ACC (%)	83.09 ¹	82.44 ± 0.65	81.51 ± 1.92	82.81 ¹	82.40 ¹
	AEOD	0.0525 ¹	0.0829 ± 0.0351	0.0829 ± 0.0470	0.0275 ¹	0.0551 ¹
	ASPD	0.0654 ¹	0.1356 ± 0.0443	0.0667 ± 0.0436	0.0488 ¹	0.0666 ¹
FedDNA	ACC (%)	83.50 ¹	82.82 ± 0.65	83.09 ± 0.76	83.76 ¹	83.72 ¹
	AEOD	0.0006 ¹	0.0882 ± 0.0722	0.0435 ± 0.0482	0.0167 ¹	0.0025 ¹
	ASPD	0.1242 ¹	0.1494 ± 0.0213	0.1178 ± 0.0204	0.1220 ¹	0.1322 ¹
FairGuard	ACC (%)	83.94 ¹	82.80 ± 0.88	62.20 ± 26.19	75.43 ¹	83.47 ¹
	AEOD	0.0006 ¹	0.0221 ± 0.0218	0.0217 ± 0.0542	0.0000 ¹	0.0014 ¹
	ASPD	0.1220 ¹	0.1089 ± 0.0272	0.0290 ± 0.0512	0.0000 ¹	0.0693 ¹
FairGuard → FLTrust	ACC (%)	83.88 ¹	82.22 ± 1.91	82.67 ± 1.81	83.75 ¹	84.08 ¹
	AEOD	0.0036 ¹	0.0141 ± 0.0150	0.0397 ± 0.0509	0.0037 ¹	0.0025 ¹
	ASPD	0.0831 ¹	0.0961 ± 0.0442	0.1102 ± 0.0431	0.0901 ¹	0.0998 ¹
Cosine + fairness [‡]	ACC (%)	82.56 ¹	81.76 ± 1.57	82.85 ± 0.83	82.71 ¹	82.56 ¹
	AEOD	0.0002 ¹	0.0226 ± 0.0177	0.0409 ± 0.0371	0.0107 ¹	0.0082 ¹
	ASPD	0.0416 ¹	0.0774 ± 0.0285	0.1110 ± 0.0125	0.0469 ¹	0.0425 ¹

Superscript 1: single seed, SD unavailable. Other cells: mean ± sample SD (ddof=1), n=10 except FedWA/FedSA n=3. No accuracy-based suppression of fairness values. AEOD denotes the implemented absolute TPR gap.

[†] FedWA is a historical display label whose numeric source is AdaAggRL. [‡] The Cosine row combines GuardFed-AD2 in Benign/S-DFA/Sp-DFA with GuardFed in F-Flip/FedSA. These are source identifications, not certifications of the named baseline algorithms.

Historical n=1 cells independently maximize ACC and minimize AEOD/ASPD over rounds61–70; 94/98 conditions have no common selected round. F-Flip/FedSA n=10 use opposite selection directions for different method groups. FedWA/FedSA n=3 jointly select ACC-.5(AEOD+ASPD). These test-selected results do not provide an unbiased matched-method ranking.

FairGuard IID FedSA ACC is restored as54.13% from the source, versus59.13% in the submitted PDF. Original submission retained separately.